

Draft email to the institute SPOC

Please read and approve, or mark anything you want changed

To: pujasarkar@iimmumbai.ac.in
Cc: vansh.260163@iimmumbai.ac.in, jinshri.260130@iimmumbai.ac.in,
Attach: SIH26171-Project-Report.pdf

Only the Cc addresses are outstanding. The email body is final.

Subject: SIH 2026 nomination: Aavaran, a privacy-preserving browser agent for ISRO (PS SIH26171)

Respected Ma'am,

I am Harsh Bajpai, BS-DSBM 2026-30, roll number 260125. I am writing on behalf of six students of the institute who have built a working entry for Smart India Hackathon 2026, and who would like to be nominated for it.

The problem. ISRO's problem statement SIH26171 asks for a browser agent that can read a user's screen and act on it without that screen being sent to a server. The obstacle is that an agent which cannot see a field cannot act on it either, so blurring the screen disables the agent, and redacting at the server concedes the point, because the data has already crossed the network by then.

What we built. Aavaran, a Chrome and Firefox extension in which a small model runs inside the browser, reads the page, and replaces every PAN, Aadhaar number, card number, password and face with a typed tag before any network request is made. An open-weight vision-language model then reasons over the censored page and returns one action, which the extension validates and executes. The tag carries the only fact the reasoning needs, that the field is filled and well formed, and withholds the value itself. The personal values never leave the machine at all, so the guarantee comes from where the data sits rather than from trusting whoever runs the server.

Where it stands. It runs today on Chrome and Firefox across Windows, macOS and Linux. Against a held-out set of pages it was never tuned on, it redacts every personal value with no false positives and leaks nothing. It completes a real multi-step task, filling and submitting a three-field government-style form, in about half a minute end to end. Seventeen checks run on every push across Windows, macOS and Linux, and every number in the attached report can be reproduced from the repository with one command.

The report sets out the architecture, the results against ISRO's own published marking scheme, and, just as importantly, the defects we found in our own work and the claims we have not yet proved.

What we are asking. We would be grateful if the institute would consider nominating our team for SIH26171; the portal closes on 30 September. We would gladly prepare whatever the institute needs from us for that, and would be happy to demonstrate the system to you in person beforehand, including on a laptop with no internet connection, which is the quickest way to see that the privacy claim is structural rather than a matter of trust.

Thank you for your time, Ma'am.

Regards,

Harsh Bajpai

BS-DSBM 2026-30 · 260125 · 9926749541

harsh.260125@iimmumbai.ac.in

On behalf of: Vansh Khosla, Manas Bharadia, Jinshri Jain, Aarna Chauhan and Siddhartha Chaudhary.

Why it is written this way

It is a pitch, not a set of questions. With no internal hackathon and only a handful of teams likely to come forward, there is nothing to ask permission about: the useful thing is to give her everything she needs to say yes, in one reading.

Three choices worth keeping if you edit it:

- **The ask is single and concrete.** One sentence, one verb, one deadline. A message

that asks for several things invites a reply that addresses none of them.

- **The bolded run-in headings carry the argument on their own.** Someone skimming four

lines gets problem, solution, status, ask, in that order.

- **The offer to demonstrate offline is the strongest line in the email.** It is the one

claim a sceptical reader can verify in two minutes, and offering it unprompted signals that the claim will survive being checked.

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